NET Physics Formula and Trick Book

1. Measurements, Units, Dimensions

Base Quantities

$$\text{Length} = L, \quad \text{Mass} = M, \quad \text{Time} = T$$
$$\text{Current} = A, \quad \text{Temperature} = K, \quad \text{Amount} = mol$$

Dimensional Formulae

$$[v] = LT^{-1}$$
$$[a] = LT^{-2}$$
$$[F] = MLT^{-2}$$
$$[p] = MLT^{-1}$$
$$[W] = [E] = ML^2T^{-2}$$
$$[P] = ML^2T^{-3}$$
$$[\rho] = ML^{-3}$$
$$[\text{pressure}] = ML^{-1}T^{-2}$$
$$[q] = AT$$
$$[V] = ML^2T^{-3}A^{-1}$$
$$[R] = ML^2T^{-3}A^{-2}$$

Common SI Units

$$F = \text{newton} = kg\,m\,s^{-2}$$
$$W, E = \text{joule} = kg\,m^2s^{-2}$$
$$P = \text{watt} = kg\,m^2s^{-3}$$
$$p = \text{momentum} = kg\,m\,s^{-1}$$
$$\text{Pressure} = Pa = N\,m^{-2}$$
$$\text{Charge} = C = A\,s$$
$$\text{Potential} = V = J\,C^{-1}$$
$$\text{Resistance} = \Omega = V\,A^{-1}$$

Errors

Absolute error:

$$\Delta x = |x_{\text{measured}} - x_{\text{true}}|$$

Relative error:

$$\frac{\Delta x}{x}$$

Percentage error:

$$\frac{\Delta x}{x} \times 100$$

For:

$$z = x^a y^b$$
$$\frac{\Delta z}{z} = |a|\frac{\Delta x}{x} + |b|\frac{\Delta y}{y}$$

Tricks

- Unit mismatch removes an option immediately.
- In multiplication/division, add percentage errors.
- In powers, multiply percentage error by power.
- Dimensional analysis cannot find numerical constants like $2, \pi, \frac{1}{2}$.
- If formula has wrong dimensions, it is definitely wrong.

2. Vectors and Equilibrium

Vector Components

$$A_x = A\cos\theta$$
$$A_y = A\sin\theta$$

Magnitude:

$$A = \sqrt{A_x^2 + A_y^2}$$

Direction:

$$\tan\theta = \frac{A_y}{A_x}$$

Resultant of Two Vectors

$$R = \sqrt{A^2 + B^2 + 2AB\cos\theta}$$

If perpendicular:

$$R = \sqrt{A^2 + B^2}$$

Dot Product

$$\vec{A}\cdot\vec{B} = AB\cos\theta$$

Perpendicular:

$$\vec{A}\cdot\vec{B} = 0$$

Cross Product

$$|\vec{A}\times\vec{B}| = AB\sin\theta$$

Parallel:

$$\vec{A}\times\vec{B} = 0$$

Equilibrium

Translational equilibrium:

$$\sum F_x = 0, \qquad \sum F_y = 0$$

Rotational equilibrium:

$$\sum \tau = 0$$

Torque:

$$\tau = rF\sin\theta$$
$$\tau = Fd_\perp$$

Lami's Theorem

For three concurrent forces in equilibrium:

$$\frac{F_1}{\sin\alpha} = \frac{F_2}{\sin\beta} = \frac{F_3}{\sin\gamma}$$

Tricks

- Resolve forces along convenient axes.
- Torque uses perpendicular distance.
- Clockwise torque equals anticlockwise torque in equilibrium.
- If object is at rest, net force is zero.
- If object moves with constant velocity, net force is also zero.

3. Motion and Force

Kinematics

$$v = u + at$$
$$s = ut + \frac{1}{2}at^2$$
$$v^2 = u^2 + 2as$$
$$s = \frac{u+v}{2}t$$

Free Fall

$$a = g$$
$$v = u + gt$$
$$h = ut + \frac{1}{2}gt^2$$
$$v^2 = u^2 + 2gh$$

Projectile Motion

Horizontal velocity:

$$v_x = u\cos\theta$$

Vertical velocity:

$$v_y = u\sin\theta$$

Time of flight:

$$T = \frac{2u\sin\theta}{g}$$

Maximum height:

$$H = \frac{u^2\sin^2\theta}{2g}$$

Range:

$$R = \frac{u^2\sin 2\theta}{g}$$

Maximum range:

$$R_{\max} = \frac{u^2}{g}$$

at:

$$\theta = 45^\circ$$

Newton's Laws

$$F = ma$$

Weight:

$$W = mg$$

Momentum:

$$p = mv$$

Impulse:

$$J = F\Delta t = \Delta p$$

Friction

$$F_f = \mu R$$

Static limiting friction:

$$F_{\max} = \mu_s R$$

Kinetic friction:

$$F_k = \mu_k R$$

Inclined Plane

Parallel component:

$$mg\sin\theta$$

Normal reaction:

$$R = mg\cos\theta$$

Acceleration down smooth incline:

$$a = g\sin\theta$$

With friction:

$$a = g(\sin\theta - \mu\cos\theta)$$

Tricks

- If time is absent, use $v^2 = u^2 + 2as$.
- If distance is absent, use $v = u + at$.
- Mass is kg; weight is newton.
- On incline, down-slope component is $mg\sin\theta$.
- Projectile horizontal velocity remains constant.
- Maximum projectile range occurs at 45°.

4. Work, Energy, Power

Work

$$W = Fs\cos\theta$$

If force and displacement are in same direction:

$$W = Fs$$

If perpendicular:

$$W = 0$$

Kinetic Energy

$$K = \frac{1}{2}mv^2$$

Potential Energy

$$U = mgh$$

Elastic Potential Energy

$$U = \frac{1}{2}kx^2$$

Work-Energy Theorem

$$W_{\text{net}} = \Delta K$$

Conservation of Energy

$$K_i + U_i = K_f + U_f$$

Power

$$P = \frac{W}{t}$$

$$P = Fv$$

Efficiency

$$\eta = \frac{\text{useful output}}{\text{input}} \times 100$$

Tricks

- Double speed gives four times kinetic energy.
- Work can be negative if force opposes displacement.
- Area under force-displacement graph is work.
- Power is rate of doing work.
- Use energy when force or acceleration is not constant.

5. Circular Motion, Rotation, Gravitation

Angular Quantities

$$\omega = \frac{\theta}{t}$$

$$\omega = 2\pi f = \frac{2\pi}{T}$$

$$v = r\omega$$

Centripetal Acceleration

$$a_c = \frac{v^2}{r} = r\omega^2$$

Centripetal Force

$$F_c = \frac{mv^2}{r} = mr\omega^2$$

Torque

$$\tau = rF\sin\theta$$

$$\tau = I\alpha$$

Rotational Kinetic Energy

$$K = \frac{1}{2}I\omega^2$$

Angular Momentum

$$L = I\omega$$

Moment of Inertia

Point mass:

$$I = mr^2$$

Gravitation

$$F = \frac{Gm_1m_2}{r^2}$$

Gravitational field:

$$g = \frac{GM}{r^2}$$

Potential:

$$V = -\frac{GM}{r}$$

Potential energy:

$$U = -\frac{GMm}{r}$$

Orbital speed:

$$v = \sqrt{\frac{GM}{r}}$$

Escape velocity:

$$v_e = \sqrt{\frac{2GM}{R}} = \sqrt{2gR}$$

Tricks

- Centripetal force is toward the center.
- Centripetal force is not a separate force; it is the net inward force.
- Doubling radius in $F = Gm_1m_2/r^2$ makes force one-fourth.
- Torque is maximum at 90°.
- If radius increases, orbital speed decreases.

6. Fluid Dynamics

Density

$$\rho = \frac{m}{V}$$

Pressure

$$P = \frac{F}{A}$$

Hydrostatic pressure:

$$P = \rho gh$$

Total pressure:

$$P = P_0 + \rho gh$$

Pascal's Principle

$$\frac{F_1}{A_1} = \frac{F_2}{A_2}$$

Buoyancy

$$F_B = \rho_{\text{fluid}} g V_{\text{displaced}}$$

Floating condition:

$$F_B = W$$

Continuity Equation

$$A_1 v_1 = A_2 v_2$$

Bernoulli's Equation

$$P + \frac{1}{2}\rho v^2 + \rho g h = \text{constant}$$

Viscosity

Stokes' law:

$$F = 6\pi\eta r v$$

Terminal velocity:

$$v_t = \frac{2r^2(\rho_s - \rho_f)g}{9\eta}$$

Surface Tension

$$T = \frac{F}{L}$$

Tricks

- Pressure in liquid increases with depth.
- Narrow pipe means higher speed.
- Higher speed means lower pressure.
- Buoyant force equals weight of displaced fluid.
- Floating body: buoyant force equals weight.

7. Elasticity

Stress

$$\text{Stress} = \frac{F}{A}$$

Strain

$$\text{Strain} = \frac{\Delta L}{L}$$

Young's Modulus

$$Y = \frac{\text{stress}}{\text{strain}}$$

$$Y = \frac{FL}{A\Delta L}$$

Bulk Modulus

$$K = -\frac{\Delta P}{\Delta V/V}$$

Shear Modulus

$$G = \frac{\text{shear stress}}{\text{shear strain}}$$

Hooke's Law

$$F = kx$$

Elastic energy:

$$U = \frac{1}{2}kx^2$$

Tricks

- Stress has unit N/m^2.
- Strain has no unit.
- Larger Young's modulus means stiffer material.
- Hooke's law is valid only within elastic limit.

8. Oscillations and SHM

SHM Condition

$$a = -\omega^2 x$$

$$F = -kx$$

Angular Frequency

$$\omega = 2\pi f = \frac{2\pi}{T}$$

Displacement

$$x = A\sin(\omega t + \phi)$$

Velocity:

$$v = \omega\sqrt{A^2 - x^2}$$

Maximum velocity:

$$v_{\max} = \omega A$$

Acceleration:

$$a = -\omega^2 x$$

Maximum acceleration:

$$a_{\max} = \omega^2 A$$

Spring-Mass System

$$T = 2\pi\sqrt{\frac{m}{k}}$$

$$f = \frac{1}{2\pi}\sqrt{\frac{k}{m}}$$

Simple Pendulum

$$T = 2\pi\sqrt{\frac{l}{g}}$$

$$f = \frac{1}{2\pi}\sqrt{\frac{g}{l}}$$

Energy in SHM

$$E = \frac{1}{2}kA^2$$

$$K = \frac{1}{2}k(A^2 - x^2)$$

$$U = \frac{1}{2}kx^2$$

Tricks

- Pendulum period does not depend on mass.
- Spring period increases if mass increases.
- At mean position: speed maximum, acceleration zero.
- At extreme position: speed zero, acceleration maximum.
- SHM acceleration is always directed toward mean position.

9. Waves and Sound

Basic Wave Relation

$$v = f\lambda$$

$$f = \frac{1}{T}$$

$$\omega = 2\pi f$$

$$k = \frac{2\pi}{\lambda}$$

Wave Equation

$$y = A\sin(\omega t - kx)$$

Speed on String

$$v = \sqrt{\frac{T}{\mu}}$$

Speed of Sound in Gas

$$v = \sqrt{\frac{\gamma P}{\rho}}$$

Intensity

$$I = \frac{P}{A}$$

For spherical wave:

$$I \propto \frac{1}{r^2}$$

Beats

$$f_b = |f_1 - f_2|$$

Doppler Effect

General idea:

$$f' = \frac{v \pm v_o}{v \mp v_s} f$$

Observer moving toward source: frequency increases.
Source moving toward observer: frequency increases.

Stationary Waves

String fixed at both ends:

$$f_n = \frac{nv}{2L}$$

Open pipe:

$$f_n = \frac{nv}{2L}$$

Closed pipe:

$$f_n = \frac{(2n-1)v}{4L}$$

Tricks

- If speed is constant, frequency and wavelength are inverse.
- Higher frequency means shorter wavelength.
- Beat frequency is difference of frequencies.
- Closed pipe has only odd harmonics.
- Intensity follows inverse square law.

10. Physical Optics

Interference

Constructive:

$$\Delta x = n\lambda$$

Destructive:

$$\Delta x = \left(n + \frac{1}{2}\right)\lambda$$

Young's Double Slit

Fringe spacing:

$$\beta = \frac{\lambda D}{d}$$

Diffraction Grating

$$d\sin\theta = n\lambda$$

Polarization

Malus law:

$$I = I_0 \cos^2\theta$$

Brewster's Law

$$\tan\theta_B = \mu$$

Resolving Power

Microscope:

$$R.P. = \frac{2\mu\sin\theta}{\lambda}$$

Tricks

- Interference requires coherent sources.
- Fringe spacing increases if wavelength increases.
- Fringe spacing decreases if slit separation increases.
- Polarization proves light is transverse.
- At Brewster angle, reflected and refracted rays are perpendicular.

11. Geometrical Optics and Instruments

Reflection

$$i = r$$

Refraction

Snell's law:

$$n_1 \sin i = n_2 \sin r$$

Refractive index:

$$n = \frac{c}{v}$$

Critical Angle

$$\sin C = \frac{1}{n}$$

for denser to rarer medium.

Lens / Mirror Formula

$$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$$

Magnification:

$$m = \frac{v}{u}$$

Lens power:

$$P = \frac{1}{f}$$

where f is in meters.

Combination of Lenses

$$P = P_1 + P_2 + \cdots$$

Simple Microscope

$$M = 1 + \frac{D}{f}$$

Telescope

$$M = \frac{f_o}{f_e}$$

Tricks

- Convert focal length from cm to m before using power.
- Convex lens is converging; concave lens is diverging.
- Total internal reflection occurs from denser to rarer medium.
- Power of convex lens is positive; concave lens is negative.
- In telescope, objective has large focal length.

12. Heat and Thermodynamics

Heat

$$Q = mc\Delta T$$

Latent heat:

$$Q = mL$$

Thermal Expansion

Linear:

$$\Delta L = \alpha L \Delta T$$

Area:

$$\Delta A = 2\alpha A \Delta T$$

Volume:

$$\Delta V = 3\alpha V \Delta T$$

Gas Laws

Boyle:

$$PV = \text{constant}$$

Charles:

$$\frac{V}{T} = \text{constant}$$

Pressure law:

$$\frac{P}{T} = \text{constant}$$

Ideal gas:

$$PV = nRT$$

Also:

$$PV = NkT$$

Kinetic Theory

Average translational K.E.:

$$K_{\text{avg}} = \frac{3}{2}kT$$

RMS speed:

$$v_{\text{rms}} = \sqrt{\frac{3RT}{M}}$$

First Law

$$\Delta Q = \Delta U + W$$

Work done by gas:

$$W = P\Delta V$$

Thermodynamic Processes

Isothermal:

$$\Delta T = 0, \quad \Delta U = 0$$

$$Q = W$$

Adiabatic:

$$Q = 0$$

$$PV^\gamma = \text{constant}$$

Isochoric:

$$\Delta V = 0, \quad W = 0$$

Isobaric:

$$P = \text{constant}$$

Efficiency

$$\eta = \frac{W_{\text{out}}}{Q_{\text{in}}}$$

Carnot efficiency:

$$\eta = 1 - \frac{T_c}{T_h}$$

Tricks

- Gas law temperature must be in Kelvin.
- During phase change, temperature remains constant.
- Constant temperature means $PV = \text{constant}$.
- For isothermal ideal gas, internal energy change is zero.
- Carnot efficiency uses Kelvin temperatures.

13. Electrostatics

Coulomb's Law

$$F = \frac{kq_1q_2}{r^2}$$

$$k = \frac{1}{4\pi\epsilon_0}$$

Electric Field

$$E = \frac{F}{q}$$

$$E = \frac{kQ}{r^2}$$

Electric Potential

$$V = \frac{W}{q}$$

$$V = \frac{kQ}{r}$$

Potential Energy

$$U = \frac{kq_1q_2}{r}$$

Relation Between Field and Potential

$$E = -\frac{dV}{dr}$$

For uniform field:

$$E = \frac{V}{d}$$

Electric Flux

$$\Phi = EA\cos\theta$$

Gauss law:

$$\Phi = \frac{q}{\epsilon_0}$$

Tricks

- Electric field is force per unit positive charge.
- Electric potential is scalar.
- Electric field is vector.
- Double distance in Coulomb's law makes force one-fourth.
- Field lines point away from positive and toward negative charge.

14. Capacitors

Capacitance

$$C = \frac{Q}{V}$$

Parallel plate capacitor:

$$C = \frac{\epsilon_0 A}{d}$$

With dielectric:

$$C = K\frac{\epsilon_0 A}{d}$$

Energy Stored

$$U = \frac{1}{2}CV^2$$

$$U = \frac{1}{2}QV$$

$$U = \frac{Q^2}{2C}$$

Capacitors in Series

$$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} + \cdots$$

For two capacitors:

$$C = \frac{C_1C_2}{C_1 + C_2}$$

Capacitors in Parallel

$$C = C_1 + C_2 + \cdots$$

Tricks

- Capacitors behave opposite to resistors.
- Capacitors in parallel add directly.
- Capacitors in series use reciprocals.
- In series, charge is same.
- In parallel, voltage is same.

15. Current Electricity

Current

$$I = \frac{Q}{t}$$

Ohm's Law

$$V = IR$$

Resistance

$$R = \rho \frac{L}{A}$$

Conductance:

$$G = \frac{1}{R}$$

Conductivity:

$$\sigma = \frac{1}{\rho}$$

Power

$$P = VI$$

$$P = I^2 R$$

$$P = \frac{V^2}{R}$$

Electrical Energy

$$E = Pt$$

$$E = VIt$$

$$E = I^2 Rt$$

Series Resistance

$$R_s = R_1 + R_2 + R_3 + \cdots$$

Same current.

Parallel Resistance

$$\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} + \cdots$$

For two:

$$R_p = \frac{R_1 R_2}{R_1 + R_2}$$

Same voltage.

Cell with Internal Resistance

$$V = E - Ir$$

Current:

$$I = \frac{E}{R + r}$$

Kirchhoff's Laws

Junction rule:

$$\sum I_{\text{in}} = \sum I_{\text{out}}$$

Loop rule:

$$\sum V = 0$$

Tricks

- Series: same current, voltage divides.
- Parallel: same voltage, current divides.
- More parallel branches reduce equivalent resistance.
- Bulb brightness depends on power.
- Internal resistance lowers terminal voltage.

16. Magnetism and Electromagnetism

Magnetic Force on Moving Charge

$$F = qvB \sin\theta$$

For circular motion in magnetic field:

$$qvB = \frac{mv^2}{r}$$

$$r = \frac{mv}{qB}$$

Force on Current-Carrying Wire

$$F = BIL \sin\theta$$

Magnetic Field Around Long Wire

$$B = \frac{\mu_0 I}{2\pi r}$$

Magnetic Field in Solenoid

$$B = \mu_0 n I$$

Torque on Current Loop

$$\tau = NBIA \sin\theta$$

Magnetic Flux

$$\Phi = BA \cos\theta$$

Tricks

- Magnetic force is zero if charge is stationary.
- Magnetic force is maximum when $v \perp B$.
- Magnetic force is zero when $v \parallel B$.
- Magnetic force does no work because it is perpendicular to velocity.
- Use right-hand rule for direction.

17. Electromagnetic Induction

Faraday's Law

$$\mathcal{E} = -\frac{d\Phi}{dt}$$

For N turns:

$$\mathcal{E} = -N\frac{d\Phi}{dt}$$

Motional EMF

$$\mathcal{E} = B\ell v$$

when rod moves perpendicular to field.

Flux

$$\Phi = BA\cos\theta$$

Self Inductance

$$\mathcal{E} = -L\frac{dI}{dt}$$

Energy in inductor:

$$U = \frac{1}{2}LI^2$$

Mutual Inductance

$$\mathcal{E}_s = -M\frac{dI_p}{dt}$$

Lenz's Law

Induced current opposes the change causing it.

Tricks

- Changing flux produces emf.
- No flux change means no induced emf.
- More turns means larger induced emf.
- Lenz's law gives direction.
- Induced current opposes change, not the field itself.

18. Alternating Current

AC Voltage and Current

$$V = V_0 \sin\omega t$$
$$I = I_0 \sin\omega t$$

RMS Values

$$V_{\text{rms}} = \frac{V_0}{\sqrt{2}}$$
$$I_{\text{rms}} = \frac{I_0}{\sqrt{2}}$$

Reactance

Inductive reactance:

$$X_L = \omega L$$

Capacitive reactance:

$$X_C = \frac{1}{\omega C}$$

Impedance

For series RLC:

$$Z = \sqrt{R^2 + (X_L - X_C)^2}$$

Current:

$$I_{\text{rms}} = \frac{V_{\text{rms}}}{Z}$$

Resonance

$$X_L = X_C$$
$$\omega_0 = \frac{1}{\sqrt{LC}}$$
$$f_0 = \frac{1}{2\pi\sqrt{LC}}$$

Power

$$P = V_{\text{rms}} I_{\text{rms}} \cos\phi$$

Power factor:

$$\cos\phi = \frac{R}{Z}$$

Transformer

$$\frac{V_s}{V_p} = \frac{N_s}{N_p}$$
$$V_p I_p = V_s I_s$$

for ideal transformer.

Tricks

- RMS value is peak divided by $\sqrt{2}$.
- In pure inductor, current lags voltage by $90°$.
- In pure capacitor, current leads voltage by $90°$.
- At resonance, impedance is minimum and current is maximum.
- Step-up transformer increases voltage but decreases current.

19. Solid State Physics

Hooke-Type Elasticity

$$Y = \frac{FL}{A\Delta L}$$

Energy Bands

Conductors: overlapping valence and conduction bands.
Insulators: large forbidden gap.
Semiconductors: small forbidden gap.

Intrinsic Semiconductor

Pure semiconductor.

Extrinsic Semiconductor

n-type:

$$\text{majority carriers} = \text{electrons}$$

p-type:

$$\text{majority carriers} = \text{holes}$$

Temperature Effect

In metals:

$$R \text{ increases with temperature}$$

In semiconductors:

$$R \text{ decreases with temperature}$$

Tricks

- n-type has electrons as majority carriers.
- p-type has holes as majority carriers.
- Semiconductor resistance decreases with temperature.
- Conductor resistance increases with temperature.

20. Electronics

Diode

Forward bias:

$$\text{conducts}$$

Reverse bias:

$$\text{almost no current}$$

Rectification

Half-wave rectifier uses one diode.
Full-wave rectifier uses two or four diodes.

Transistor Currents

$$I_E = I_B + I_C$$

Current gain:

$$\beta = \frac{I_C}{I_B}$$

$$\alpha = \frac{I_C}{I_E}$$

Relation:

$$\beta = \frac{\alpha}{1-\alpha}$$

$$\alpha = \frac{\beta}{1+\beta}$$

Logic Gates

AND:

$$Y = A \cdot B$$

OR:

$$Y = A + B$$

NOT:

$$Y = \overline{A}$$

NAND:

$$Y = \overline{AB}$$

NOR:

$$Y = \overline{A+B}$$

Tricks

- Diode conducts in forward bias.
- Reverse bias widens depletion region.
- Transistor has $I_E = I_B + I_C$.
- Small base current controls large collector current.
- NAND and NOR are universal gates.

21. Dawn of Modern Physics

Photon Energy

$$E = hf$$

$$c = f\lambda$$

$$E = \frac{hc}{\lambda}$$

Photoelectric Effect

$$hf = \phi + K_{\max}$$

$$K_{\max} = hf - \phi$$

$$eV_0 = K_{\max}$$

Threshold frequency:

$$f_0 = \frac{\phi}{h}$$

Threshold wavelength:

$$\lambda_0 = \frac{hc}{\phi}$$

de Broglie Wavelength

$$\lambda = \frac{h}{p}$$
$$p = mv$$
$$\lambda = \frac{h}{mv}$$

Relativity

Mass-energy:

$$E = mc^2$$

Energy change:

$$\Delta E = \Delta mc^2$$

Compton Effect

$$\Delta\lambda = \frac{h}{m_e c}(1 - \cos\theta)$$

Tricks

- Larger frequency means larger photon energy.
- Larger wavelength means smaller photon energy.
- Photoelectric effect depends on frequency, not intensity.
- Intensity increases number of emitted electrons, not their maximum energy.
- de Broglie wavelength decreases as momentum increases.

22. Atomic Spectra

Bohr Radius

$$r_n = n^2 r_1$$

Energy Level

$$E_n = -\frac{13.6}{n^2}\ \text{eV}$$

Energy Difference

$$\Delta E = E_f - E_i$$

Photon emitted/absorbed:

$$hf = |E_i - E_f|$$

Rydberg Formula

$$\frac{1}{\lambda} = R\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$$

where:

$$n_2 > n_1$$

Series

Lyman:

$$n_1 = 1$$

Balmer:

$$n_1 = 2$$

Paschen:

$$n_1 = 3$$

Tricks

- Electron falling to lower level emits photon.
- Electron moving to higher level absorbs photon.
- Lyman series lies in ultraviolet.
- Balmer series lies in visible region.
- Higher n means less negative energy.

23. Nuclear Physics

Nuclear Radius

$$R = R_0 A^{1/3}$$

Mass Defect

$$\Delta m = Zm_p + (A - Z)m_n - m_{\text{nucleus}}$$

Binding Energy

$$E_b = \Delta mc^2$$

Binding Energy Per Nucleon

$$\frac{E_b}{A}$$

Radioactive Decay

$$N = N_0 e^{-\lambda t}$$

Activity:

$$A = \lambda N$$

Half-life:

$$T_{1/2} = \frac{0.693}{\lambda}$$

After n half-lives:

$$N = N_0 \left(\frac{1}{2}\right)^n$$
$$n = \frac{t}{T_{1/2}}$$

Alpha, Beta, Gamma

Alpha:

$${}^4_2He$$

Beta-minus:

$$n \to p + e^- + \bar{\nu}$$

Gamma:

high energy photon

Fission and Fusion

Fission: heavy nucleus splits.
Fusion: light nuclei combine.

Tricks

- After 1 half-life, remaining amount is 1/2.
- After 2 half-lives, remaining amount is 1/4.
- After 3 half-lives, remaining amount is 1/8.
- Alpha emission decreases mass number by 4 and atomic number by 2.
- Gamma emission changes neither mass number nor atomic number.
- Fusion powers the sun.

24. Quick Physics MCQ Elimination Rules

Unit Rules

- Force: $kg\,m\,s^{-2}$.
- Energy: $kg\,m^2 s^{-2}$.
- Power: $kg\,m^2 s^{-3}$.
- Pressure: N/m^2.
- Charge: $A\,s$.
- Potential: J/C.
- Magnetic field: $N/(A\,m)$.

Graph Rules

- Slope of displacement-time graph = velocity.
- Slope of velocity-time graph = acceleration.
- Area under velocity-time graph = displacement.
- Area under force-displacement graph = work.
- Area under power-time graph = energy.

Proportional Reasoning

$$F \propto \frac{1}{r^2}$$

Doubling r makes F one-fourth.

$$K \propto v^2$$

Doubling v makes kinetic energy four times.

$$P \propto V^2$$

for constant resistance.

$$v = f\lambda$$

If v constant, f and λ are inverse.

Most Common Traps

- Mixing mass and weight.
- Forgetting Kelvin conversion.
- Using cm instead of m in lens power.
- Treating centripetal force as outward.
- Forgetting square in inverse-square laws.
- Confusing series and parallel rules.
- Forgetting RMS value is peak divided by $\sqrt{2}$.
- Thinking intensity changes photoelectron energy.
- Using $Q = mc\Delta T$ during phase change instead of $Q = mL$.
- Forgetting magnetic force is zero when motion is parallel to field.

Fast Attempt Method

1. Identify chapter/topic.
2. Write governing formula.
3. Check units.
4. Check proportional relation.
5. Substitute only if needed.
6. Compare option magnitude and sign.
7. Mark the trap in one line.